

LogiTwin

AI Engine

Full Product Documentation

Predictive B2B Supply Chain Simulation and Automated Mitigation System

4	23	81	<2s	6.75
Subsystems	API Endpoints	Tests Passing	Detection	Max Impact Score

Table of Contents

01	What is LogiTwin?	The problem, the solution, who it's for
02	How It Works	The 5-step pipeline from ingest to mitigation
03	The Four Subsystems	SS1 Ingestion, SS2 Simulation, SS3 Mitigation, SS4 Dashboard
04	Graph Database Architecture	Why Neo4j, the schema, Cypher queries
05	Impact Scoring Formula	How factories are ranked by disruption severity
06	API Reference	All 23 endpoints with descriptions
07	Authentication	JWT tokens, roles, security
08	Real-Time Streaming	Server-Sent Events, dashboard integration
09	Industries Supported	Automotive, pharma, electronics, FMCG
10	Deployment Options	SaaS pilot, on-premise, hybrid
11	Test Coverage	81 unit tests across all subsystems
12	Getting Started	How to request a pilot

01 — What is LogiTwin?

LogiTwin AI Engine is a predictive B2B supply chain simulation and automated mitigation system. It models your supply chain — every supplier, factory, and port — as a live graph database, detects disruptions in real time, scores the downstream impact on every factory, and automatically fires carrier API hooks to reroute shipments, activate backup suppliers, and notify procurement teams.

The Problem

- Supply chain disruptions cost the global economy \$4 trillion annually. Most companies have no system to detect which nodes are affected, score the downstream impact, and automatically trigger rerouting in real time.
- SAP, Oracle, and legacy tools run on relational databases built in the 1990s — before graph databases existed. They cannot traverse a Tier 1/2/3 supply chain in milliseconds. They generate alerts — not actions.
- By the time a company knows Hamburg port is congested and BMW's supply chain is at CRITICAL risk, the production line has already stopped. LogiTwin detects and acts in under 2 seconds.

The Solution

- Graph-native architecture — Neo4j stores every supply chain node and relationship. BFS traversal finds all affected factories in milliseconds, not hours.
- Real-time detection — Celery async workers process disruption events, traverse the graph, score factories, and write back to Neo4j — all in under 2 seconds.
- Automated mitigation — HTTPX async client fires carrier API hooks in parallel with retry, backoff, and full exception isolation. One failed hook never blocks others.

Who is LogiTwin for?

Role	What LogiTwin does for them
Supply Chain Head	See every disruption risk in real time. Know which factories are affected before your team does. Execute mitigations from the dashboard.
CTO / VP Engineering	Evaluate a production-grade, fully tested, open-source-stack supply chain intelligence engine. Full API docs at /docs. 81 tests passing.
CEO / Investor	LogiTwin detects BMW-level CRITICAL disruptions in 2 seconds at 0 infrastructure cost. That's the product-market fit in one sentence.
Procurement Manager	Get ranked mitigation options with ETA delta and cost delta for every disruption. Click Execute — hook fires to carrier API automatically.
Logistics Manager	Real-time dashboard with live factory capacity, port status, and disruption event feed. SSE stream updates every 3 seconds.

02 — How It Works — The 5-Step Pipeline

Step	What Happens	Technology	Output
1. Ingest	Suppliers, factories, and ports are loaded into Neo4j via REST API endpoints with strict Pydantic v2 validation. MERGE-based upserts prevent duplicates.	FastAPI + Pydantic v2 + Neo4j	Graph nodes and relationships in Neo4j
2. Detect	A disruption event is fired — port_congestion, supplier_offline, capacity_degradation, or weather_strike. FastAPI queues the task immediately and returns a task_id.	FastAPI + Celery + Redis	task_id returned instantly — non-blocking
3. Traverse	Celery worker creates its own Neo4j driver and runs BFS Cypher queries to find every downstream factory — via direct SUPPLIES paths AND maritime SHIPS_THROUGH routes simultaneously.	Neo4j Cypher BFS + async driver	List of all affected factories with transit data
4. Score	Impact formula applied to each factory: severity_input x max(transit_days,1) x max(1.0-capacity_rate, 0.05). Results ranked CRITICAL/HIGH/MEDIUM/LOW. Written to Neo4j as DISRUPTION_EVENT node with IMPACTS relationships.	scorer.py + Neo4j write	Ranked factory impact list written to graph
5. Mitigate	Hook registry maps disrupted node to carrier API endpoints. HTTPX async client fires hooks in parallel — asyncio.gather(return_exceptions=True). Results written as MITIGATION_ACTION relationships. Event status updated to mitigated.	HTTPX + hook_registry + Neo4j	Carrier APIs called, audit trail in graph

03 — The Four Subsystems

Subsystem	Name	Files	Endpoints	Tests
SS1	FastAPI Ingestion Engine	config.py, models.py, db.py, main.py, services/graph.py, routes/ingest.py	8	—
SS2	Stochastic Simulation Engine	workers/celery_app.py, workers/simulation.py, services/traversal.py, services/scorer.py, routes/simulation.py	5	21
SS3	Automated Mitigation Layer	workers/hook_registry.py, services/mitigation.py, routes/mitigation.py, models.py additions	4	27
SS4	Productization and Demo Layer	auth/jwt_handler.py, auth/dependencies.py, routes/auth.py, sse.py, dashboard/app.py, scripts/	6	33
TOTAL	Full LogiTwin AI Engine	20+ Python files	23	81

Key Architectural Decisions

Decision	Reason
Celery workers own their Neo4j driver	Workers are separate OS processes. Sharing a driver causes race conditions. Each worker creates/closes its own driver per task.
asyncio.gather(return_exceptions=True)	One failed carrier hook never cancels others. Every connector is independently isolated.
MERGE-based Neo4j upserts	Re-ingesting the same supplier 100 times never creates duplicates. Idempotent by design.
Pydantic v2 strict mode	Malformed data rejected at the API boundary with clear error messages — before touching the database.
Starlette 0.41.3 pinned	Newer versions break FastAPI router init in this environment. Documented and tested.

04 — Graph Database Architecture

LogiTwin uses Neo4j — the world's leading graph database. Supply chains are naturally graphs: nodes are suppliers, factories, and ports; edges are the delivery routes between them. Neo4j traverses this graph in milliseconds for queries that would take seconds in SQL.

Node Types

Node Label	Properties	What it represents
(:Supplier)	id, name, tier (1/2/3), location	A company that provides components — TSMC, Bosch, Tata Steel, MRF Tyres
(:Factory)	id, name, capacity_rate (0.0-1.0), location	An assembly plant — Maruti Suzuki Gurugram, BMW Chennai, Toyota Bangalore
(:Port)	id, name, average_clearance_days, status	A maritime port — JNPT Mumbai, Hamburg, Yokohama, Busan
(:DISRUPTION_EVENT)	id, disruption_type, severity_input, highest_risk_level, status, timestamp, mitigated_at	A detected disruption — created by SS2 simulation engine on every run

Relationship Types

Relationship	Properties	Meaning
[SUPPLIES]	component_id, lead_time_days	Supplier delivers component directly to factory. Lead time in days.
[SHIPS_THROUGH]	transit_days	Supplier ships goods to a port. Transit days for this leg.
[DELIVERS_TO]	transit_days	Port delivers goods to factory. Transit days for this leg.
[IMPACTS]	impact_score, risk_level, path_type	Written by SS2 — connects DISRUPTION_EVENT to each affected factory.
[MITIGATION_ACTION]	action_id, carrier_name, success, duration_ms, executed_at, error	Written by SS3 — full audit trail of every mitigation hook fired.

05 — Impact Scoring Formula

Every affected factory is scored using a deterministic formula that captures three dimensions of disruption impact simultaneously:

```
impact_score = severity_input x max(transit_days, 1) x max(1.0 - capacity_rate, 0.05)
```

Where:

severity_input = 0.0 (minor) to 1.0 (complete failure) – from the disruption event

transit_days = total days in the disrupted supply path (direct or maritime)

capacity_rate = factory's current capacity (0.0-1.0). 70% capacity = 0.70

(1-capacity_rate) = factory vulnerability. BMW at 70% = 0.30 vulnerability

max(...,0.05) = floor ensures even 100% capacity factories score non-zero

Risk Level Thresholds

Risk Level	Score Range	Real Example	Action
CRITICAL	5.0 or above	BMW India 6.75 — Hamburg 90% blocked, 24-day path, 70% capacity	Immediate automated mitigation
HIGH	2.0 to 4.99	Honda India 4.60 — Renesas 95% offline, 22-day path, 78% capacity	Urgent — trigger SS3 hooks
MEDIUM	0.5 to 1.99	Maruti Suzuki 1.94 — multi-path disruption, 92% capacity reduces exposure	Monitor — queue options
LOW	Under 0.5	Direct factory hit — no supply chain cascade — transit_days = 0	Log and watch

06 — API Reference — All 23 Endpoints

Full interactive documentation is available at /docs (Swagger UI) when the API is running. All endpoints require JWT Bearer token authentication except /health and /auth/token.

Ingestion Endpoints (SS1)

Method	Path	Description
POST	/ingest/supplier	Add supplier node. MERGE-based — safe to re-ingest.
GET	/ingest/suppliers	List all supplier nodes in graph.
POST	/ingest/factory	Add factory with capacity_rate and location.
GET	/ingest/factories	List all factories with capacity rates.
POST	/ingest/port	Add port with clearance days and status.
GET	/ingest/ports	List all ports with status and clearance times.
POST	/ingest/relationship/supplies	Create SUPPLIES relationship — supplier to factory.
POST	/ingest/relationship/ships-through	Create maritime path — supplier to port to factory.

Simulation Endpoints (SS2)

Method	Path	Description
POST	/simulation/run	Fire custom disruption. Returns task_id immediately (non-blocking).
GET	/simulation/status/{task_id}	Poll result. Returns full impact report when status=SUCCESS.
POST	/simulation/inject/{1,2,3}	Inject predefined mock scenario for quick demos.
GET	/simulation/events	List all DISRUPTION_EVENT nodes, newest first.
GET	/simulation/scenarios	List available mock scenarios for injection.

Mitigation Endpoints (SS3)

Method	Path	Description
GET	/mitigation/options/{event_id}	Ranked reroute candidates — hook registry + graph paths combined.
POST	/mitigation/execute/{event_id}	Fire selected hook. dry_run=true for testing without real API call.

Method	Path	Description
GET	/mitigation/hooks	Full hook registry — all nodes, all carriers, all actions.
GET	/mitigation/events	Events with mitigation status — pending, mitigated, or failed.

Auth and Streaming Endpoints (SS4)

Method	Path	Description
POST	/auth/token	Login — returns JWT Bearer token valid 60 minutes.
GET	/auth/me	Current user info from token claims.
GET	/auth/verify	Token validity check — 200 if valid, 401 if expired.
GET	/stream/disruptions	SSE stream — pushes live disruption data every 3 seconds.
GET	/stream/health	SSE health ping — heartbeat every 5 seconds.
GET	/health	API and database health check.

07 — Authentication

LogiTwin uses JWT (JSON Web Token) authentication with HS256 signing. Two roles exist — admin (full access) and viewer (read-only).

Getting a Token

```
POST /auth/token
Content-Type: application/x-www-form-urlencoded
username=admin&password=logitwin-admin-2024

Response:
{"access_token":"eyJhbGciOi...", "token_type":"bearer", "expires_in":3600, "role":"admin"}
```

Using the Token

```
GET /ingest/suppliers
Authorization: Bearer eyJhbGciOi...

All endpoints (except /health and /auth/token) require this header.
Token expires after 60 minutes — request a new one via /auth/token.
```

Roles

Role	Access	Credentials
admin	Full access — ingest, simulate, mitigate, execute hooks, view all data	admin / logitwin-admin-2024
viewer	Read-only — view events, options, hooks, graph data. Cannot ingest or execute.	viewer / logitwin-viewer-2024

08 — Industries Supported

Industry	Key Disruption Types	LogiTwin Models	Live Demo?
Automotive	Semiconductor shortage, port blockage, factory capacity degradation	31 suppliers, 8 factories, 6 ports — Indian OEMs fully seeded	YES — live
Pharmaceuticals	API ingredient shortage, cold chain port disruption, regulatory closure	API ingredient suppliers, cold chain ports, manufacturing sites	Available on request
Electronics	Fab capacity, rare earth shortage, geopolitical port risk	Semiconductor supply trees — TSMC, Samsung, rare earth Tier 3	Available on request
FMCG / Retail	Packaging shortage, warehouse overload, co-packer disruption	Packaging suppliers, contract manufacturers, distribution hubs	Available on request

09 — Deployment Options

Option	Description	Setup Time	Best For
Free Pilot (SaaS)	LogiTwin hosted on our infrastructure. We seed your supply chain. You access via API and dashboard.	1 week	First evaluation — zero setup required
On-Premise	We deploy LogiTwin on your servers. Neo4j, Redis, FastAPI all on your infrastructure. Data never leaves your network.	2-4 weeks	Enterprise with data sovereignty requirements
Hybrid	Neo4j graph on-premise. Simulation engine and dashboard in cloud. Best performance with data control.	3-5 weeks	Mid-size enterprise with partial cloud adoption

Tech Stack Requirements

Component	Technology	Version	Purpose
API Server	Python + FastAPI + Uvicorn	Python 3.11+	REST API, async request handling
Graph Database	Neo4j	5.20.0	Supply chain graph storage and traversal
Task Queue	Celery + Redis	Celery 5.4, Redis 7	Async simulation workers
HTTP Client	HTTPX	0.28.1	Carrier API hook execution
Dashboard	Streamlit	1.45+	Real-time supply chain monitoring UI
Container	Docker	Latest	Neo4j and Redis containerization
Auth	python-jose	3.5.0	JWT HS256 token creation and verification

10 — Test Coverage — 81 Tests Passing

Test File	Suite	Tests	What is covered	Result
test_simulation.py	TestCalculateImpactScore	6	Formula correctness, edge cases, known values, rounding	6/6 PASS
test_simulation.py	TestClassifyRiskLevel	4	CRITICAL/HIGH/MEDIUM/LOW boundary values	4/4 PASS
test_simulation.py	TestScoreAffectedFactories	4	Empty results, single, multiple factories, required keys	4/4 PASS
test_simulation.py	TestMockFeeds	7	All scenarios load, invalid raises, shortcut injectors	7/7 PASS
test_mitigation.py	TestHookRegistry	12	All nodes have hooks, priorities sorted, lookup by ID, empty node	12/12 PASS
test_mitigation.py	TestHookClientDryRun	5	Dry run success, payload echo, parallel fire, exception isolation	5/5 PASS
test_mitigation.py	TestRecommendationBuilder	6	Urgency text, carrier names, severity thresholds	6/6 PASS
test_mitigation.py	TestMitigationModels	4	Pydantic v2 instantiation, defaults, validation	4/4 PASS
test_auth.py	TestTokenCreation	9	JWT format, claims, expiry, different users, custom expiry	9/9 PASS
test_auth.py	TestTokenVerification	8	Valid, invalid, tampered, wrong secret, missing sub, expired	8/8 PASS
test_auth.py	TestUserAuthentication	10	Admin, viewer, wrong password, unknown user, case sensitive	10/10 PASS
test_auth.py	TestUsersDB	6	Required fields, roles, non-empty passwords	6/6 PASS
TOTAL	All Subsystems	81	Pure unit tests — zero mocks of core business logic	81/81 PASS

11 — Get Started — Request Your Free Pilot

LogiTwin offers a free 30-day pilot for qualified supply chain organizations. We model your actual network, run live disruption simulations, and deliver a complete impact report. No commitment required.

Step	Activity	Timeline
1	Submit pilot request via shivang-csa.github.io/logitwin/contact.html or email shivangwork1121@gmail.com	Day 0
2	30-minute data collection call — we gather your supplier list, factory locations, and port dependencies	Day 1-2
3	Graph seeding — your supply chain loaded into LogiTwin Neo4j with all Tier 1/2/3 relationships	Day 3-7
4	Simulation runs — 5 disruption scenarios based on your actual past incidents or highest-risk scenarios	Day 7-21
5	Demo day — live LogiTwin dashboard walkthrough with your supply chain team. See real scores in real time.	Day 28-30
6	Decision — if LogiTwin adds value, we discuss pricing. If not, you keep the full impact report. No commitment.	Day 30

Contact: shivangwork1121@gmail.com | Website: shivang-csa.github.io/logitwin | Free 30-day pilot — no commitment